

Table 1: A comparison of IoT simulation tools

Features	COOJA	NS-3	OMNeT++	NetSim
Simulation purpose	Wireless sensor networks	Network and IoT	Network and IoT	Network and IoT
Licence	BSD	Open source	Academic public licence	Open source
Simulation language	Java	C++ and Python	C, C++	Java, HTML and XML
Mobility support	Yes	Yes	Yes	Yes
Customisation	Limited	Extensible	Highly extensible	Extensible
Network protocols	IEEE 802.15.4, 6LoWPAN	Various IoT protocols	Various IoT protocols	Various IoT protocols
Operating systems	Cross-platform	Cross-platform	Cross-platform	Windows-based
Ease of use	Beginner-friendly	Requires coding skills	Requires coding skills	Beginner-friendly with scripting
Real-time simulation	Limited	Supported	Supported	Limited support
Application areas	WSN and IoT research	A broad range of IoT apps	IoT and communication	IoT research and development

Table 2: Popular Python libraries for IoT-enabled healthcare projects

Library	Description	Features	Use cases	Version
OpenCV	Computer vision and image processing	Image, video processing, object detection, image recognition	Medical imaging, diagnostic analysis, patient monitoring	OpenCV 4.8.0
PyHealth	Machine learning library for healthcare	Simplifies data preprocessing and model deployment	Predictive analytics, patient risk stratification, readmission prediction	PyHealth 1.1.4
MNE-Python	Data analysis for electrophysiology and neuroimaging	Analysis of EEG, MEG, and other neural data, machine learning tools for data interpretation	Neuroimaging research, epilepsy monitoring, brain-computer interfaces	mne 1.5. 1
BlueZ	Bluetooth stack for IoT and healthcare devices	Supports IoT devices' Bluetooth connectivity	Wearable health monitors, Bluetooth-enabled medical devices	BlueZ 5.63
PyDICOM	DICOM file manipulation for medical imaging	Read and write DICOM files – helps extract and manipulate metadata from medical images	Medical image management, radiology, picture archiving, and communication systems (PACS)	PyDICOM 1.1.0
scikit-learn	Machine learning for healthcare data analysis	Supports classification, regression, clustering	Predictive modelling, disease prediction, patient risk assessment	scikit-learn 1.3.1